Conjecture 00000004279: infinitely many rank-three Butler groups

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1 The finite-isomorphism-quotient clause is false

For every prime p take the actual localization and additive group

$$R_p = \mathbf{Z}[1/p], \qquad G_p = R_p \oplus R_p \oplus R_p.$$

These are rank-three Butler groups, and $G_p \ncong G_q$ whenever $p \neq q$. Since there are infinitely many primes, the set of isomorphism classes of all rank-three Butler groups is infinite. This disproves the source's necessary claim that the quotient is finite up to isomorphism. It does not deny a classification with an infinite parameter space or decide a modified problem fixing the rank-one types.

We follow the precise English definition in the source: a finite-rank pure subgroup of a direct sum of rank-one groups. The standard torsion-free formulation is recalled in Fuchs and Rangaswamy, *Valuated Butler groups of finite rank*, J. Aust. Math. Soc. 80 (2006), Introduction, printed p.336, item (1), before the authors introduce the valued version. Our examples are already completely decomposable; their realization as pure subgroups uses the whole ambient direct sum, not a requirement that the subgroup be proper.

2 Actual localization, rank, and purity

Lean uses Mathlib's actual localization `Localization.Away` of $\mathbf{Z}$ at p. The rank-one assertion is the genuine module rank

$$\operatorname{rank}_{\mathbf{Z}} R_p = 1,$$

obtained from the rank-of-localization theorem. The canonical map $R_p \to \mathbf{Q}$ is constructed by the localization universal property and proved injective, so each summand is torsion free.

The group G_p is represented first by the finite function space $\operatorname{Fin} 3 \to R_p$. Its rank is not assigned by definition. The actual injective $\mathbf{Z}$-linear maps

$$\mathbf{Z}^3 \longrightarrow G_p \longrightarrow \mathbf{Q}^3$$

give lower and upper bounds of three on its module rank. The upper bound uses the rank-preserving passage from $\mathbf{Z}$ to its fraction field $\mathbf{Q}$. Consequently

$$\operatorname{rank}_{\mathbf{Z}} G_p = 3.$$

No freeness of R_p as a $\mathbf{Z}$-module is assumed. Injectivity of the rational map also proves torsion freeness of G_p.

Lean explicitly supplies the linear equivalence between this finite function space and the actual direct sum $D_p = \bigoplus_{i \in \mathrm{Fin}\, 3} R_p$. For an actual integer submodule $H \leq D_p$, the purity condition is encoded by the usual scalar-equation criterion: for every natural n and $h \in H$, if $nb = h$ has a solution $b \in D_p$, then it has a solution in H. This is exactly $nD_p \cap H = nH$. The full submodule $H = D_p$ satisfies that criterion immediately. Thus the formal witness consists of actual rank-one torsion-free summands, an actual finite direct sum, an actual pure submodule and a linear isomorphism from the actual rank-three torsion-free group. The theorem `family_qualifies` collects these properties.

3 An additive-group invariant separates the primes

Multiplication by p is surjective on R_p because p is invertible in this localization. It is therefore surjective on all three coordinates of G_p.

If $q \neq p$ is another prime, the image of p is a unit in $\mathbf{Z}/q\mathbf{Z}$. The universal property gives a genuine ring homomorphism

$$\rho_{p,q} : R_p \longrightarrow \mathbf{Z}/q\mathbf{Z}.$$

If multiplication by q were surjective on G_p, apply it to the vector $(1, 1, 1)$. Its first coordinate would give $qx = 1$ in R_p. Applying $\rho_{p,q}$ gives $0 = 1$ in the nonzero field $\mathbf{Z}/q\mathbf{Z}$, a contradiction. Hence

$$G_p \text{ is } p\text{-divisible, but is not } q\text{-divisible for } q \neq p.$$

Surjectivity of multiplication by a fixed natural number is preserved by every additive-group isomorphism. It follows that no additive isomorphism $G_p \cong G_q$ exists for distinct primes. This obstruction applies to all additive-group isomorphisms without requiring preservation of the chosen ring presentation.

4 The real isomorphism quotient is infinite

On the actual type of primes Lean defines the relation

$$p \sim q \quad \Longleftrightarrow \quad \mathrm{Nonempty}(G_p \simeq_+ G_q).$$

It checks reflexivity, symmetry and transitivity using actual additive equivalences. Pairwise nonisomorphism makes the map from primes to this quotient injective, so the quotient is infinite. A separate theorem excludes every finite complete catalog: any assignment of labels to these groups for which equal labels imply isomorphism would inject the infinite set of primes into a finite type. The relation is not defined as equality of the prime labels.

5 Verification

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. The package includes a new root build, transitive axiom audits and full official `leanchecker --fresh Main` replay. There is no `sorry`, `native_decide` or extra axiom. OpenAI Codex assisted; a separate supervising Codex agent reviewed source, report and PDF. This is AI review, not human certification. Organizer acceptance is pending.